CHAPTER 1 · MASTER 1 COMPUTER SCIENCE

Modelling & Simulation

Play with the world without breaking it 🎮



Pr. Latifa DEKHICI · latifa.dekhici@univ-usto.dz · USTO-MB

💪 WARM-UP

Which would you hate to test live?

🚑 **A new ER triage rule**

💻 **A new firewall rule**

🏭 **A new factory layout**

Discuss with your neighbor: what could go wrong in each case?

Why not test on the real thing?

🏥 Healthcare

Close half the ER beds on a busy Friday to see how long patients wait? Nobody should be the guinea pig.

🏭 Manufacturing

Stop a car-factory line just to try a new setup? Every minute of silence costs a fortune.

💻 IT

Crash a live data center to test the backup servers? Hello, angry users.

Fix: build a model, a safe copy of reality, and experiment on that ✨

QUICK VOTE

Impossible, harmful or impractical?

1

Moving the Moon to study the tides.

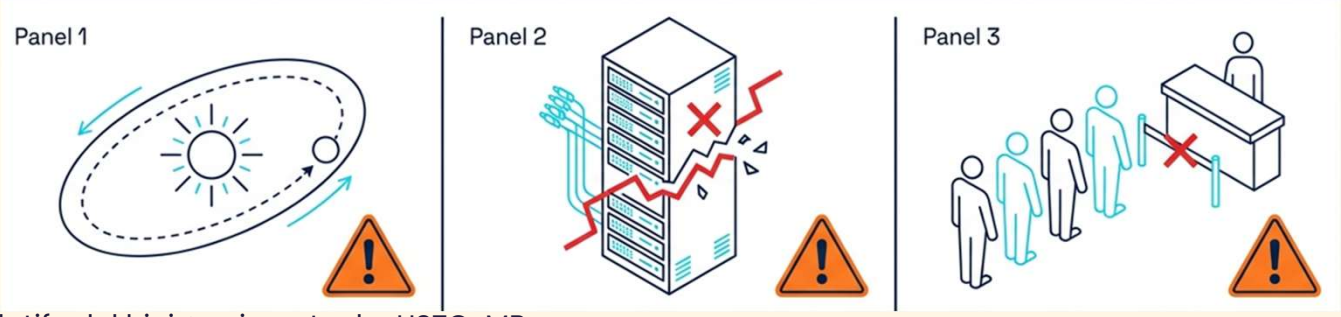
2

Switching off a hospital database to test the backups.

3

Cutting cashiers at rush hour to watch the queue.

Answers: 1 Impossible · 2 Harmful · 3 Impractical (customers would leave for good)



Models are caricatures 🎨

They drop the boring details to keep the mechanism that matters.

George E.P. Box

"All models are wrong, but some are useful."

Jay W. Forrester

"Models are not reality. They are tools for helping us understand reality."

Albert Einstein

"As far as the laws of mathematics refer to reality, they are not certain."

A model is a metro map 🚇

It skips the streets and buildings and keeps only what helps you get around.

“All models are wrong, but some are useful.”
George E.P. Box, statistician

Our focus: dynamic + discrete + stochastic, like queues in ERs, factories and routers.

What is a system?

Things that work together for one goal. Example: an emergency room.

1 · Components

👩‍⚕️ Doctors, nurses, patients, beds.

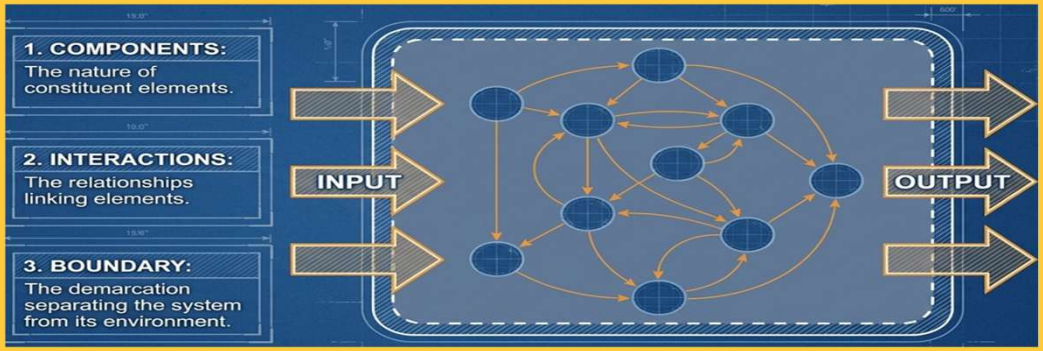
2 · Interactions

A patient goes to triage, then to a doctor, then to a bed.

3 · Boundary

The ER walls. The ambulance outside is the environment.

Where you draw the line depends on your question: one ER, or the whole city?



🙋 **YOUR TURN**

Draw the system: a pharmacy

1 · Components

Name 3 of them.

2 · Interactions

Name 2 of them.

3 · Boundary

Is the delivery truck inside or outside?

Example: pharmacists, shelves, patients · a patient hands over a prescription · the truck is outside, so it is environment.

Five words, three worlds

Word	🏥 ER	🏭 Factory	💻 Router
Entity (who or what)	Patient	Bottle on the line	Data packet
Attribute (its details)	Urgency level	Size, color	Size, priority
Activity (takes time)	Treating a patient	Filling a bottle	Forwarding a packet
Event (instant change)	Patient arrives	Bottle reaches machine	Packet arrives
State (the situation now)	Patients waiting	Bottles waiting	Packets in queue



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🙋 YOUR TURN · 5 MIN · IN GROUPS

Fill in the table



Word	🍩 Donut shop	🏦 Bank
Entity	?	?
Attribute	?	?
Activity	?	?
Event	?	?
State	?	?



Possible answers ✅

Word	🍩 Donut shop	🏦 Bank
Entity	Customers, counter	Customers, tellers
Attribute	Order size, priority	Account balance, type
Activity	Serving a customer	Making a deposit
Event	Customer arrives or leaves	Customer arrives, service ends
State	People in the queue	Busy tellers, queue length

Where do events come from?

Inside the system

- 🏥 A doctor finishes with a patient.
- 🏭 A machine finishes a part.
- 💻 A router forwards a packet.

Outside the system

- 🚑 An ambulance shows up.
- 📦 A big new order arrives.
- 🌐 A flood of users hits the website.

Inside = endogenous. Outside = exogenous.

🙋 QUICK VOTE · 3 MIN

Inside or outside the system?

1

A machine finishes a part.

2

A power cut hits the data center.

3

A nurse finishes treating a patient.

4

A new patient walks in.

Answers: 1 inside · 2 outside · 3 inside · 4 outside

Modelling vs simulation

🗺️ Modelling

Drawing the digital version of the ER: beds, doctors, patient flow.

🚶 Simulation

Walking around that map: what if we add two nurses?

Modelling builds the representation. Simulation explores how it behaves under different conditions.

The modelling spectrum

🧪 Physical

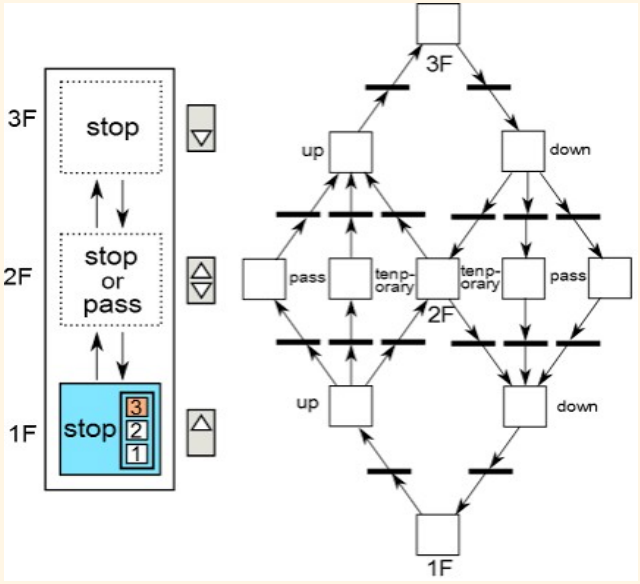
A scaled-down copy: wind tunnel, crash-test dummy, mini elevator.

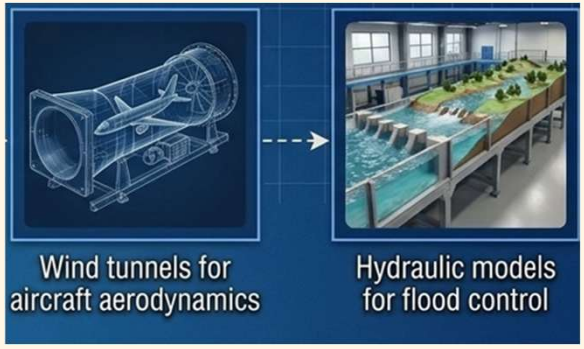
🧮 Analytical

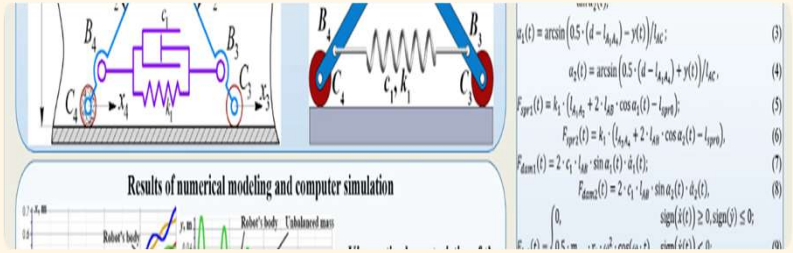
A closed formula gives the answer: distance = speed × time.

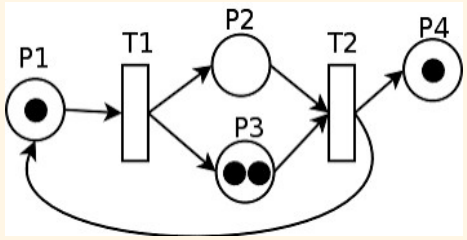
💻 Simulation

Too complex for a formula, so the computer runs it: a whole hospital.









Three ways to sort models

🎲 Sure or random?

Deterministic: 10 pizzas × 8 slices = 80 slices, every time.

Stochastic: ER arrivals. No two Fridays look alike.

📷 Photo or movie?

Static: a snapshot, like one risk estimate.

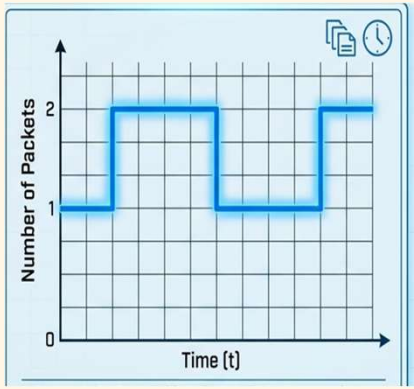
Dynamic: the system changes over time, like a waiting room.

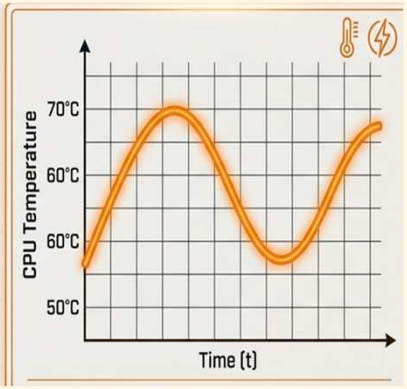
🪜 Steps or slide?

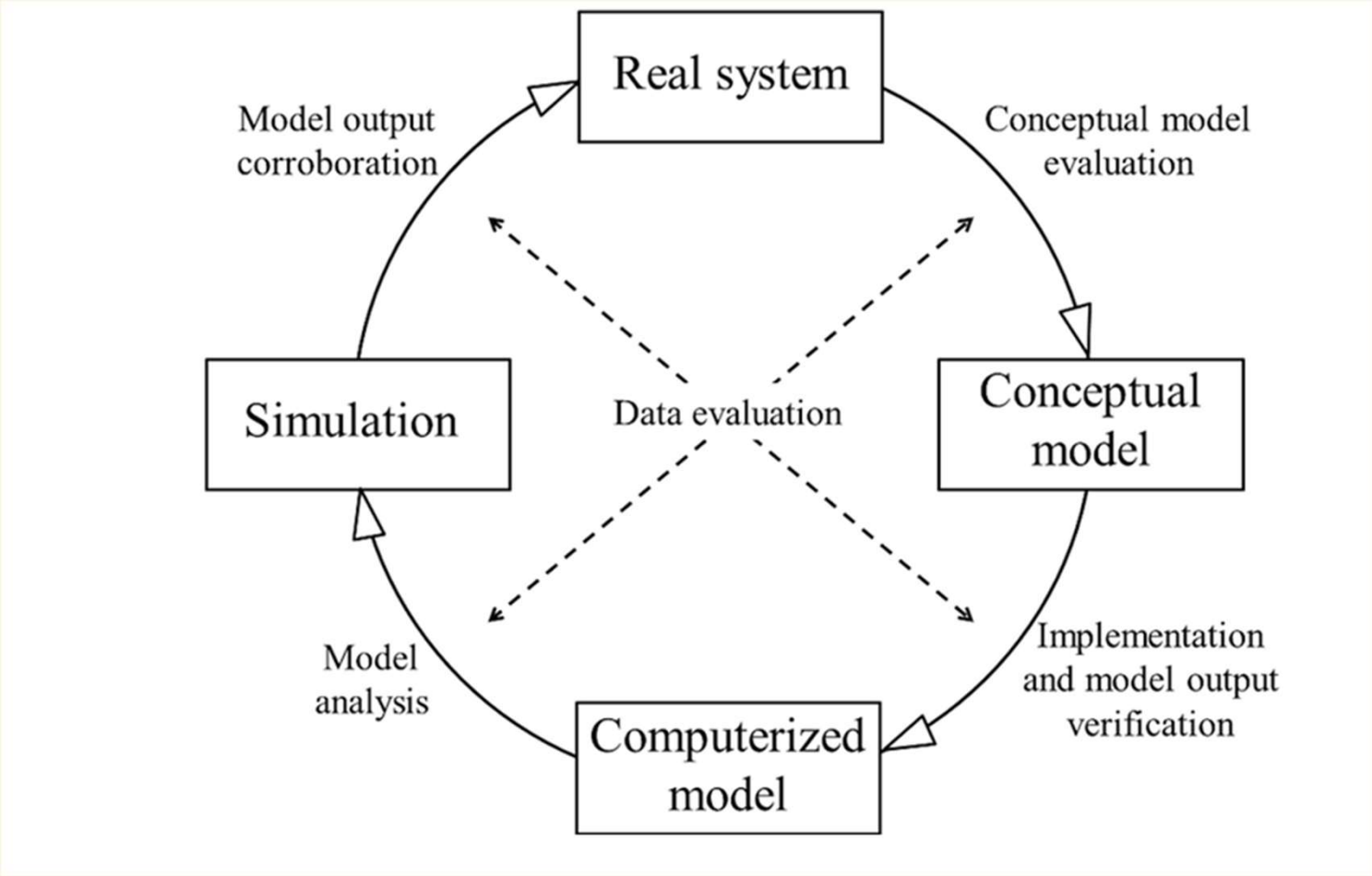
Discrete: queue jumps from 12 to 13 patients.

Continuous: machine temperature rises smoothly.








Real system
Model output corroboration
Conceptual model evaluation
Simulation
Data evaluation
Conceptual model
Model analysis
Implementation and model output verification
Computerized model

The classification matrix

Static + Discrete

Poker odds, one-time pharmacy stock order.

Dynamic + Discrete ⭐

ER queue, router buffer, factory line. Our focus.

Static + Continuous

Stress on a bridge under a fixed load.

Dynamic + Continuous

Heartbeat overnight, CPU temperature, dam water level.

🙋 **YOUR TURN •**

Which box?

1

Patients in a waiting room.

2

Server temperature over 24 hours.

3

Odds of a lottery win.

4

Fixed load on a crane arm.

Solutions·

Which box?

1

Patients in a waiting room.

2

Server temperature over 24 hours.

3

Odds of a lottery win.

4

Fixed load on a crane arm.

Answers: 1 Dynamic + Discrete · 2 Dynamic + Continuous · 3 Static + Discrete · 4 Static + Continuous

🎯 **EXIT TICKET**

Can you answer these?

1

Name the 3 pillars of a system.

2

Give a random (stochastic) example from health or IT.

3

Which matrix box do queues live in?

Answers: components, interactions, boundary · patient arrivals · Dynamic + Discrete